

# **NEE Quantum States and Population Transfer Algorithm**

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# Introduction

This is a collection of unpolished notes summarizing the main aims and results during the research internship undertaken by the author(s) in the summer of 2026, with respect to NEE states and its application in quantum population transfer (PT) algorithms.

## 1 Preliminary Calculations

This section confirms some analytical calculations for the LRP model used in the QREM proposed in [1].

The problem to be solved is as follows: given one low-energy bit string (a deep local minimum), one wishes to find other bit strings with nearly the same low energy but are far away in Hamming distance. Concisely, the input is a known low-energy marked state  $z_j$ , and the goal is to identify other marked states  $z \neq z_j$  whose energy lies within a narrow target window  $\Delta\mathcal{E}_{cl}$  around the initial  $z_j$ . The most general form of classical Hamiltonian model  $H_{cl}$  assumed in [1] is in the form

$$H_{cl} = \sum_z \mathcal{E}(z) |z\rangle \langle z| \quad (1)$$

where  $\mathcal{E}(z)$  is any classical energy function over  $2^n$  bit strings  $z = (z_1, z_2, \dots, z_n)$  with  $z_k = \{0, 1\}$ . Here the main qualitative assumptions are:

- (i) The energy landscape of  $\mathcal{E}_{cl}$  has many local minima.
- (ii) These low-energy minima are separated by substantial Hamming distances (at least  $d \sim O(n)$ ).
- (iii) The number of relevant low-energy states is exponential in  $n$  yet exponentially small compared to  $2^n$  (which makes it computationally difficult).

Actually, a more specific classical Hamiltonian (partly inspired by the impurity band model in doped semiconductors) is described in [1] which allows more detailed analysis:

$$H_{cl} = \sum_{j=1}^M \mathcal{E}(z_j) |z_j\rangle \langle z_j| \quad (2)$$

for  $M \gg 1$ , and which is evolved under a "driver Hamiltonian" (responsible for the PT protocol) given by

$$H_D = -B_{\perp} \sum_{k=0}^n \sigma_x^k \quad (3)$$

Together we get

$$H = H_{cl} + H_D \quad (4)$$

In (2), it is further assumed that:

- (i) Only the marked states have low energy, and all the other states have  $\mathcal{E} = 0$ . These unmarked states are also separated by an enormous energy gap  $\simeq n$  from the narrow band of marked states.
- (ii) Marked states  $|z_j\rangle$  are chosen uniformly and randomly from all bit strings of length  $n$  (thus, from all  $2^n$  possible strings).
- (iii) Marked states have energies  $\mathcal{E}(z_j)$  fall independently around  $-n$  in a narrow band of width  $W \ll B_\perp$ , so are approximately given by

$$\mathcal{E}(z_j) \simeq -n + \epsilon_j \quad (5)$$

where  $|\epsilon_j| \ll W/2$ .

In [1], one works with the following PDF in the tailed probability distribution:

$$P(w) = \frac{1}{w^2 \sqrt{\pi \log w}}, \quad w \in [1, \infty), \quad (6)$$

where  $\omega_{ij} = (V_{ij}/V_{typ})^2$  is the rescaled squared amplitude. One can convert this to be in terms of  $V$ , as

$$\begin{aligned} P(V) &= P(W) \left| \frac{dw}{dV} \right| \\ &= \frac{1}{(V^2/V_{typ}^2)^2 \sqrt{\pi \ln(V^2/V_{typ}^2)}} \frac{2V}{V_{typ}^2}. \end{aligned}$$

Thus one can conclude

$$P(V) \sim \frac{V_{typ}^2}{V^3 \sqrt{\ln(V/V_{typ})}}. \quad (7)$$

Given the format for the Lévy-Rosenzweig-Porter model:

$$P_{LRP}(V_{ij}) \sim \frac{2\mu N^{-\mu\gamma/2}}{|V_{ij}|^{1+\mu}} \theta(|V_{ij}| - N^{-\gamma/2}), \quad (8)$$

note that

$$\mu = 2 + \epsilon. \quad (9)$$

where  $\epsilon$  is a small residue logarithmic correction negligible compared to unity. Hence, for the

fractal dimension  $D_q$ , we have

$$D_q = \frac{\mu(2 - \gamma)}{2(\mu - 1)} = 2 - \gamma. \quad (10)$$

Therefore, one can compare the sizes of the full bandwidth  $W$  with off-diagonal  $V_{typ}$ , to arrive at the three regimes

$$\begin{cases} W \lesssim V_{typ} M^{1/2}, & \text{Ergodic,} \\ V_{typ} M^{1/2} \ll W \ll V_{typ} M, & \text{Non-ergodic delocalized (NEE),} \\ W \gtrsim V_{typ} M, & \text{Localized.} \end{cases} \quad (11)$$

To determine the properties of the model exactly, we need to also calculate  $\gamma$  in terms of known variables. Here we note that

$$V_{typ} \sim n^2 2^{-n/2} e^{-n/(4B_\perp^2)}. \quad (12)$$

as well as referring to [1] with

$$W = \lambda M^{\gamma/2} V_{typ}, \quad (13)$$

where  $\lambda$  is just a parameter of order  $\mathcal{O}(\infty)$ . Then one can simultaneously solve them to arrive at an expression of  $\gamma$  dependent on  $M$ :

$$\gamma = \frac{2}{\ln M} \left[ \ln \left( \frac{W}{\lambda} \right) + \frac{n}{2} \ln 2 + \frac{n}{4B_\perp^2} \right]. \quad (14)$$

The total number of states  $M$  can be graphically represented as corresponding to the shaded area in *Figure 1*, assuming our marked states is around  $E = -\epsilon n$  with width  $W$ . Quantitatively, the Gaussian distribution allows us to calculate  $M$ :

$$M(E) \simeq W \cdot \text{DOS}(E) = \frac{W 2^n}{\sqrt{2\pi n \sigma^2}} \exp \left( -\frac{E^2}{2n\sigma^2} \right). \quad (15)$$

Substituting the specific value  $E = -\epsilon n$  and taking logarithm, we get

$$\ln M = n \ln 2 - \frac{\epsilon^2 n}{2\sigma^2} + \ln W - \frac{1}{2} \ln (2\pi n \sigma^2). \quad (16)$$

Hence, we substitute (16) into (14) to finally calculate  $\gamma$  in terms of just  $B_\perp$  and  $\sigma$  which are

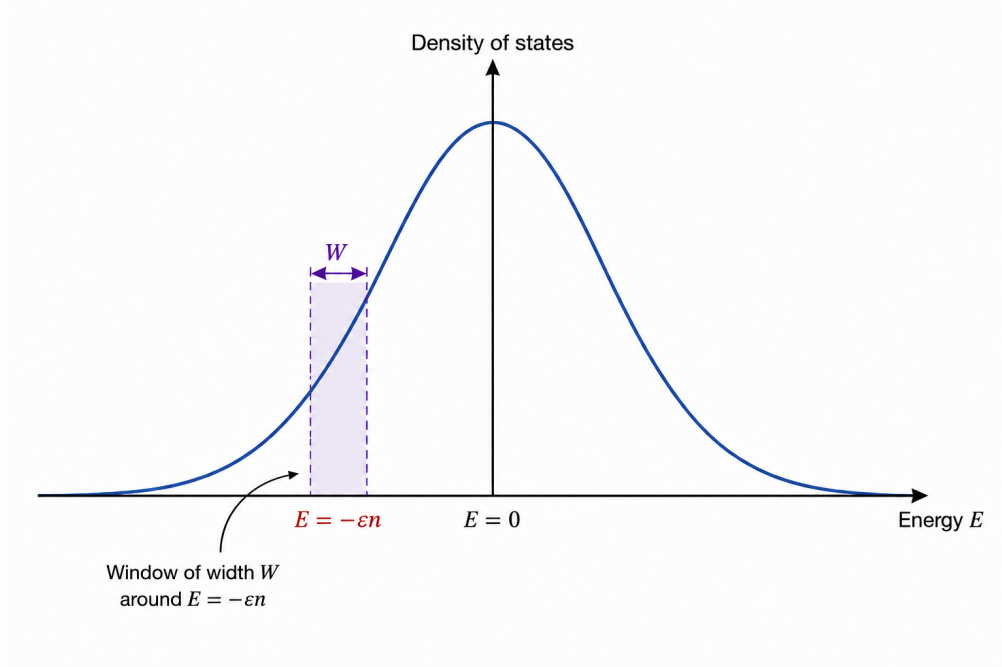


Figure 1: Density of states under a canonical Gaussian distribution (oversimplified).

intrinsic properties known to the system:

$$\gamma = \frac{\ln(2) + 1/(2B_{\perp}^2)}{\ln(2) - \epsilon^2/(2\sigma^2)}. \quad (17)$$

where we assumed the limits that  $\ln(W)/n \rightarrow 0$  and  $\ln(n)/n \rightarrow 0$  for large values of  $n \gg 1$ . This form of  $\gamma$  in (17) is in the same form as that found in [2].

## 2 Trotterization and PT Algorithm Initiative

To test whether the algorithm proposed in [1] works, one could consider a time-evolution of the initially chosen state  $|z_0\rangle$ . Breaking the total time  $t$  into  $L$  steps, with each timestep  $\Delta t = t/L$ , one could define the unitary operator through trotterization:

$$U(t) \simeq \left( e^{-iH_{cl}\Delta t} e^{-iH_D\Delta t} \right)^L, \quad (18)$$

where  $H_D$  and  $H_{cl}$  are defined as in (3) and (1). To first order, one can therefore write the final evolved state as

$$|z_0(t)\rangle = \prod_{l=1}^L e^{-iH_{cl}\Delta t} e^{-iH_D\Delta t} |z_0\rangle, \quad (19)$$

where we can then check the success rate

$$P(t) = \sum_{\substack{z \neq z_0 \\ |E(z) - E(z_0)| < \Delta E_{cl}}} |\langle z | \psi(t) \rangle|^2, \quad (20)$$

which should grow to a final value whilst leaving the original  $z_0$ . If such case is observed then this algorithm is indeed demonstrated. Additionally, the characteristic timescale  $t_p$  of the PT algorithm is given by the following:

$$t_p \sim \frac{1}{\Gamma} \sim \frac{1}{V_{typ} M^{1-\gamma/2}} \sim \frac{M^{\gamma-1}}{W}, \quad (21)$$

where  $\Gamma$  is the typical miniband width. One has to confirm  $t \sim t_{typ}$  to confirm the success of the algorithm claimed. Now we propose a brief summary of the coding initiatives to verify the trotterization process, before proceeding to test the PT algorithm:

1. Set parameters. Choose  $n = 10$ , transverse field  $\Gamma$ , target energy density  $\epsilon$ , timestep  $\Delta t$ , and total time  $t$ .
2. Generate the classical REM energies. Draw  $2^n = 1024$  independent Gaussian energies with

$$\mathcal{E}_z \sim \mathcal{N}(0, n/2) \quad (22)$$

and build  $H_{cl}$  based on (1). This is only an arbitrary Gaussian choice, as the structure of these classical energies may have effects on final results, and we will investigate whether this is crucial for the performance of the algorithm.

3. Choose the initial low-energy state. Pick the closest

$$z_0 = \arg \min_z |E_z + \epsilon n| \quad (23)$$

such that  $E_{z_0} \simeq -\epsilon n$ .

4. Build the driver Hamiltonian  $H_D$  as in (3). So together one gets (4).
5. Compute exact evolution:

$$|\psi_{exact}(t)\rangle = \exp(-iHt) |z_0\rangle. \quad (24)$$

6. Compute the evolved state from trotterization, as in (19).
7. Compare by using the fidelity

$$F(t) = |\langle z_0 | \psi(t) \rangle|^2. \quad (25)$$

After checking that for reduced  $\Delta t$  the value of  $F(t)$  is converging, one can conclude that this trotterization scheme works fine. Now for the PT algorithm, we could thus consider the success rate outlined in (20), getting the numerical results to be compared with the theory claimed in [1].

### 3 Sparse Full Matrix Quantum Evolution

To actually check the validity of the PT algorithm, we need to make some initial setup step by step and utilize numerical simulations. It is instructive to follow this section with the Jupyter notebook at hand.

Our model simulated is a bit different from our previous Gaussian shape DOS in *Figure 1*. Instead, we consider a broad Gaussian background of unmarked states and a narrow, low-energy marked band; the transverse field  $B_\perp$  then drives quantum transitions between computational-basis states. An illustrative diagram is shown in *Figure 2*.

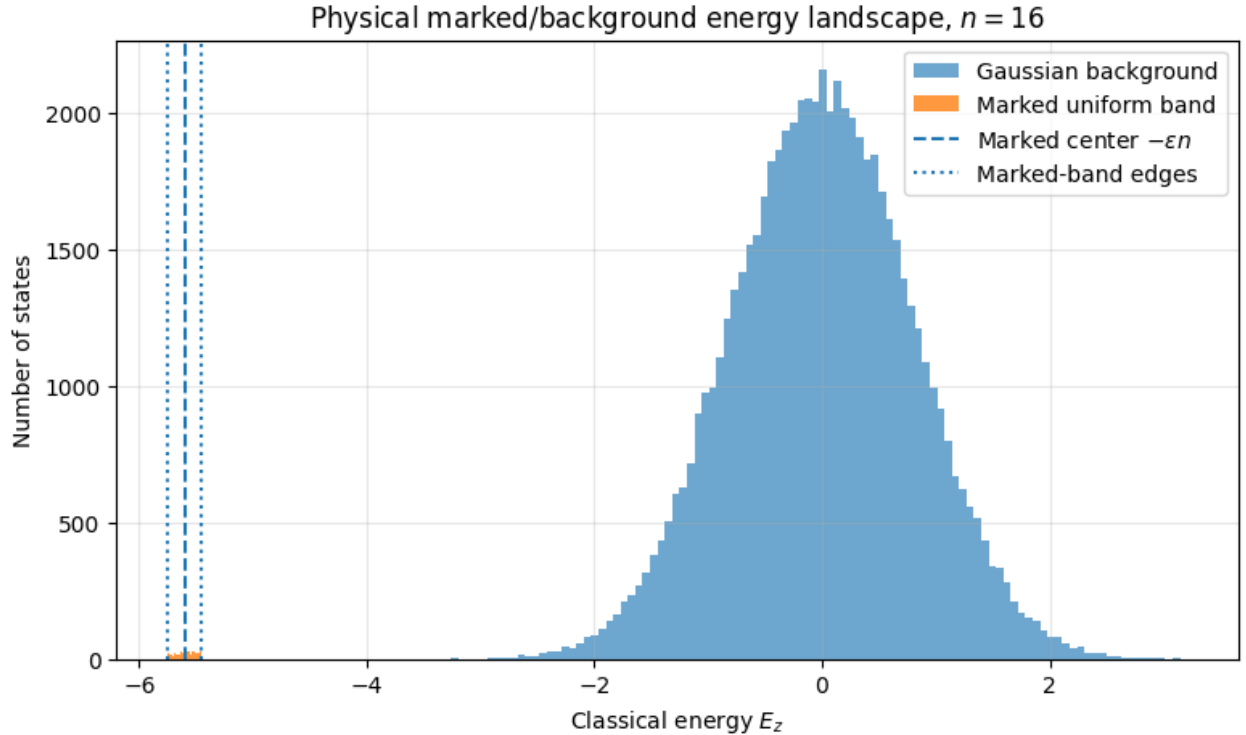


Figure 2: Density of states under the canonical model (Gaussian + uniform marked band). The numbers on the axes are arbitrary and serve as a valid example.

In particular, one has to clearly define a few key parameters involved. For the classical background, the Gaussian has a SD of  $\sigma_0$  and associated width  $\sigma_{E,0} = \sigma_0\sqrt{n}$ , whilst we also

assume an artificial broader Gaussian encompassing both the classical background and the uniform marked energy window (which is centered at  $E_{\text{marked}} = -\epsilon n$ , with width  $W$ ), with a SD of  $\sigma$  and associated width  $\sigma_E = \sigma\sqrt{n}$ . The sole purpose of the latter is to make an estimate of the marked-state density.

At this point, one has to carefully place  $\epsilon$ . Specifically, it has to lie below from the very edge of the classical Gaussian to be completely separated. For the Gaussian distribution of width  $\sigma_0\sqrt{n}$ , the probability density is approximately

$$P_G(E) \sim \exp\left(-\frac{E^2}{2\sigma_0^2 n}\right). \quad (26)$$

The edge of the exponentially populated spectrum is obtained by asking when the population becomes order one:

$$2^n \exp\left(-\frac{E^2}{2\sigma_0^2 n}\right) \sim O(1). \quad (27)$$

Writing  $E_{\text{low}} = -\delta n$ , then at leading order

$$\delta = \sigma_0\sqrt{2\ln 2}, \quad (28)$$

which specifies the lower edge of the classical background distribution. Therefore, for  $E_{\text{marked}} < E_{\text{low}}$ , we need

$$\epsilon > \sigma_0\sqrt{2\ln 2}. \quad (29)$$

Also, recall that whilst using the artificial Gaussian with width  $\sigma\sqrt{n}$  for estimating the marked band, we still require there to be exponentially many states at that position. If we write the population as

$$M(E) \sim 2^n \exp\left(-\frac{\epsilon^2 n}{2\sigma^2}\right) \sim \exp\left[n\left(\ln 2 - \frac{\epsilon^2}{2\sigma^2}\right)\right], \quad (30)$$

then we need  $\ln 2 - \epsilon^2/2\sigma^2 > 0$ , giving us

$$\epsilon < \sigma\sqrt{2\ln 2}. \quad (31)$$

Combining (29) and (31), one has the essential criterion

$$\sigma_0\sqrt{2\ln 2} < \epsilon < \sigma\sqrt{2\ln 2}. \quad (32)$$

Afterwards, we would like to define the starting state and the set of states that count as

successful PT outcomes. The initial state is the marked state closest to the center:

$$z_0 = \arg \min_i |E_i + \epsilon n|. \quad (33)$$

The target condition is set as

$$|E_i - E_{z_0}| \leq \Delta E_{\text{success}} \simeq \frac{W}{\eta}, \quad (34)$$

where  $\eta$  is some arbitrary constant number (for example  $\eta = 12$  in our numerical simulation). We can record the number of target states within this narrower band to be  $N_{\text{target}}$ , while the total number of marked states in the lower uniform band is  $M$ , estimated from the artificial Gaussian DOS with SD of  $\sigma$ . An example statistics is shown below in *Figure 3*.

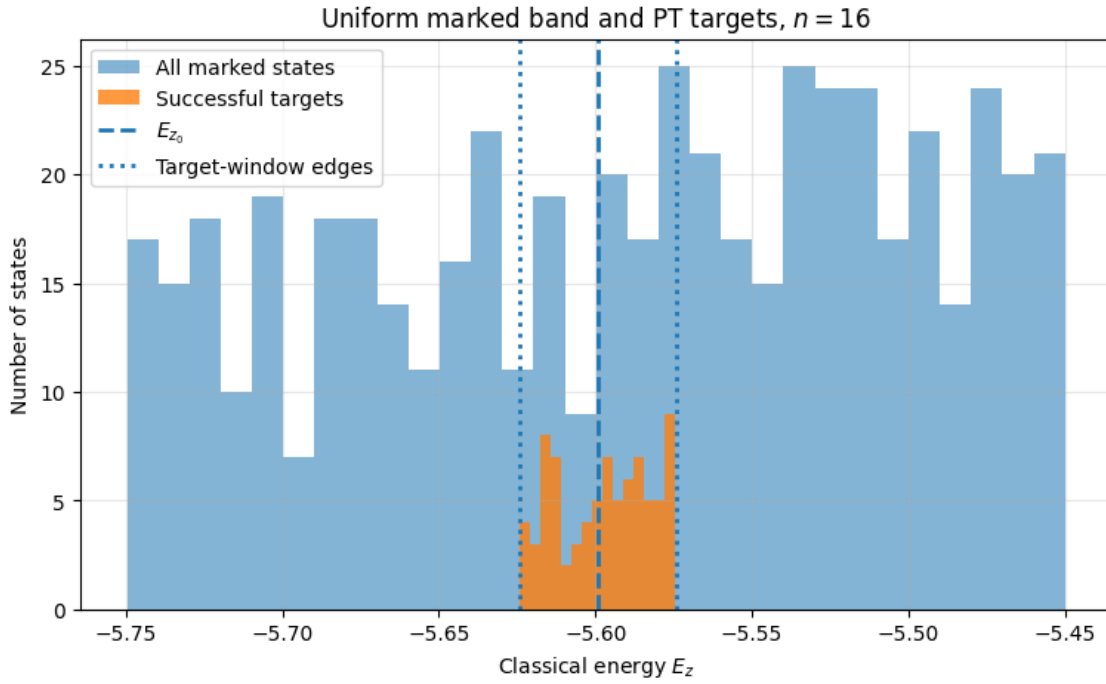


Figure 3: Target states within the energy band of marked states. The numbers on the axes are arbitrary and serve as a valid example.

Some useful quantities can be derived at this stage. We proceed by calculating the mean level spacing for marked states:

$$\delta_M = \frac{W}{M}, \quad (35)$$

and therefore the marked-sector Heisenberg time is

$$t_H^{(M)} = \frac{1}{\delta_M} = \frac{M}{W}. \quad (36)$$

The inverse band-width timescale is

$$t_W = \frac{1}{W}. \quad (37)$$

The effective miniband size (number of states)  $\Omega$  is given by [1]

$$\Omega = \left( \frac{\pi V_{typ}}{\delta_M} \right)^2, \quad (38)$$

where one can estimate  $V_{typ}$  by (12). The characteristic miniband width is given by

$$\Gamma_{typ} = V_{typ} \sqrt{\frac{\pi \Omega \ln \Omega}{4}}, \quad (39)$$

from which we know the characteristic PT timescale

$$t_{PT} \simeq t_{typ} = \frac{1}{\Gamma_{typ}}. \quad (40)$$

This is the theoretical timescale that we subsequently compare with the actual numerical evolution.

Still, before performing the PT evolution, we conduct some regime diagnostics to ensure our quantum states do live in the NEE regime, neither fully ergodic nor localized.

Condition 1:

$$V_{typ} \sqrt{M} < W < V_{typ} M. \quad (41)$$

This is just the NEE case outlined in (11).

Condition 2:

$$1 < \Omega < M. \quad (42)$$

If  $\Omega < 1$ , it would mean there is essentially no multi-state miniband.  $\Omega \sim M$  would indicate that the relevant states occupy essentially the whole marked sector, approaching ergodic behavior. Thus (42) exactly implies a proper intermediate miniband.

Condition 3:

$$\delta_M < \Gamma_{typ} < W. \quad (43)$$

This means the miniband width is: much larger than the individual marked-state level spacing, so several states participate. And it is much smaller than the entire marked-band width, so the dynamics is not spread across the full marked band.

Then we could follow the procedure of trotterization scheme as in (19), and consider the numerical evolution of PT algorithm from  $|z_0\rangle$  for  $\sim 10t_{PT}$ , for various  $n$ .

A few key quantities need to be carefully manipulated. Here the fidelity with the initial

state (overlap) is defined as

$$P_0(t) = |\langle z_0 | \psi(t) \rangle|^2. \quad (44)$$

The probability of transferring into the marked-state band is

$$P_{\text{success}}(t) = \sum_{z \in \text{target}} |\langle z | \psi(t) \rangle|^2. \quad (45)$$

The success probability (total) of the PT algorithm is

$$P_{\text{marked}}(t) = \sum_{z \in \text{marked}} |\langle z | \psi(t) \rangle|^2. \quad (46)$$

An example of the PT evolution outcomes is shown in *Figure 4*. One can see that roughly a stabilizing characteristic timescale is  $t_W = 1/W$ , which is much shorter than  $t_{PT}$ . Through calculations, it is also possible to check that  $P_{\text{marked}} \gg M/N$  and  $P_{\text{success}} \gg N_{\text{target}}/N$  where  $N = 2^n$ , giving a significant preference to select states in the marked energy band, with a proportionately very high weight associated with this region.

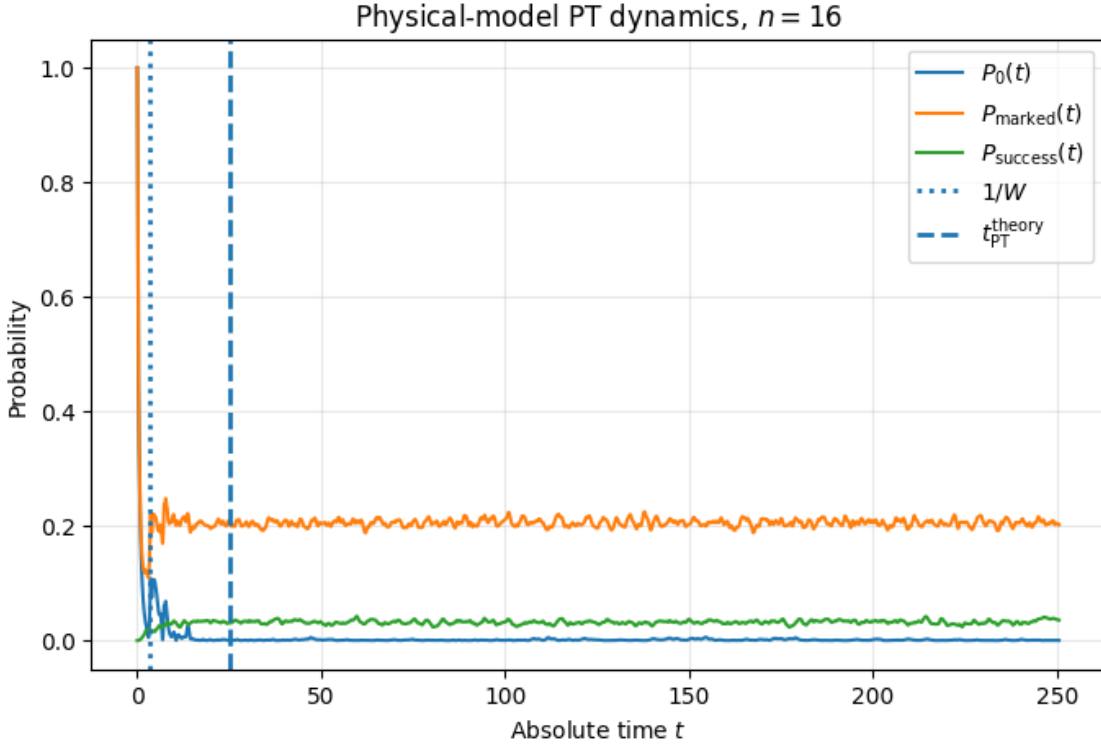


Figure 4: PT evolution outcomes for  $n = 16$ .

To quantify this preference, we define a numerical quantity known as global selectivity:

$$S_{\text{global}}(t) = \frac{P_{\text{success}}(t)/(1 - P_0(t))}{N_{\text{target}}/(N - 1)}, \quad (47)$$

where the numerator is the probability of success within all accessible states except the initial state  $z_0$ , and the denominator is the statistical probability of randomly selecting states in the targets from all non-initial states of the total ensemble (including both classical background and the lower marked band).

*Figure 5* has plotted this global selectivity scale over time, which shows a clear  $S_{\text{global}} \gg 1$  tendency for our PT evolution, for all times  $t \gg 1/W$  or  $t \gg t_{PT}$ .

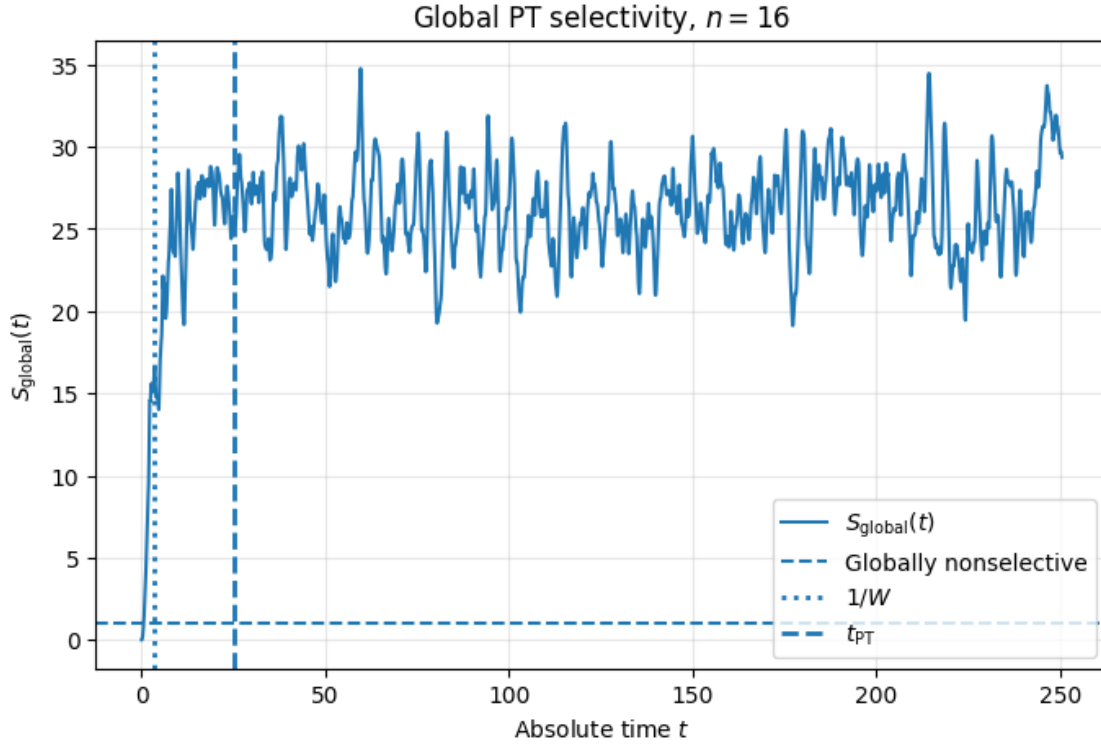


Figure 5: Global selectivity over time for  $n = 16$ .

It is also possible to define internal selectivity as

$$S_{\text{internal}}(t) = \frac{P_{\text{success}}/N_{\text{target}}}{P_{\text{non-target}}/N_{\text{non-target}}}, \quad (48)$$

which tells us the target preference within the marked sector. From our numerical evolution, the result is plotted in *Figure 6*, which shows  $S_{\text{internal}} \sim 1$  and there is no bias towards the target states in the band. This result might not correspond to the initial motive in [1] and

would require further analysis in the future.

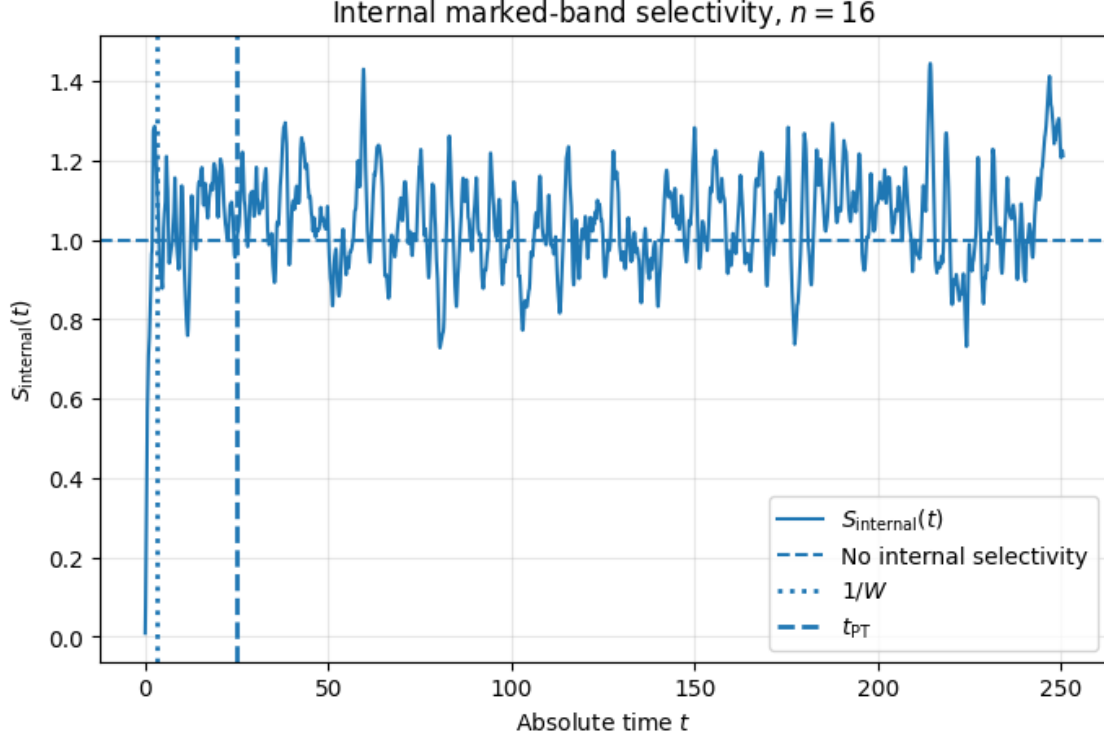


Figure 6: Internal selectivity over time for  $n = 16$ .

The entire set of  $P_{\text{success}}$  for various  $n$  values is shown in the Jupyter Notebook at the end, and one could check it to confirm the  $1/W$  timescale required to reach a steady plateau. Global and internal selectivity could also be computed for all such cases, showing more or less the same expected behavior as  $n = 16$ . One could also change the numerical values of parameters used, as long as the required criteria described beforehand could be satisfied.

## 4 Projected Dense Matrix Analysis

This is an unfinished task to check the consistency of the quantum Hamiltonian model, for which I will just briefly describe the key insights to be followed, and detailed coding could then follow.

The PT protocol is concerned with the low-energy marked configurations, so instead of retaining the complete  $2^n$  dimensional space, we identify the relevant  $M$ -dimensional marked-state subspace.

We define

$$P = \sum_{i=1}^M |z_i\rangle\langle z_i| \quad (49)$$

as the projector onto the marked states, and

$$Q = 1 - P \tag{50}$$

is the projector onto the remaining unmarked states.

Then from some quantum mechanical calculations, it is not hard to show that the down-folded dense Hamiltonian  $H_{\text{eff}}$  is

$$H_{\text{eff}}(E) = PHP + PHQ \frac{1}{E - QHQ} QHP, \tag{51}$$

where  $H$  is the full sparse Hamiltonian originally. Theoretically  $H_{\text{eff}}$  should be written in the form of

$$H_{\text{eff}} = \text{diag}(\epsilon_i) + V, \tag{52}$$

where the classical marked-state energies remain on the diagonal part and  $V_{ij}$  contains the information concerning effective tunnelling.

For the purpose of checking, we should obtain  $V_{ij}$  from (51) as the actual off-diagonal matrix elements. The paper [1] theoretically claims that a Lévy-Rosenzweig-Porter distribution is expected for  $\omega_{ij} = (V_{ij}/V_{\text{typ}})^2$  and mathematically expressed in the form of (6). This comparison between actual calculation from downfolding and theoretical claim should be made to check the validity of the model.

Finally, one should use the same  $H_{\text{eff}}$  (including its diagonal terms) to investigate whether the expected miniband structure and population-transfer behaviour follow.

## References

- [1] Vadim N. Smelyanskiy, Kostyantyn Kechedzhi, Sergio Boixo, Sergei V. Isakov, Hartmut Neven, and Boris Altshuler. Nonergodic delocalized states for efficient population transfer within a narrow band of the energy landscape. *Physical Review X*, 10(1), January 2020.
- [2] Lara Faoro, Mikhail V. Feigel'man, and Lev Ioffe. Non-ergodic extended phase of the quantum random energy model. *Annals of Physics*, 409:167916, October 2019.